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LLM Dual-Agent for Taste Preference Prediction

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***Abstract*—Taste preference prediction plays a crucial role in enhancing consumer satisfaction and driving innovation in areas such as food science and personalized nutrition. Current approaches often rely on large language models (LLMs), yet these methods remain constrained by sparse domain knowledge, fragmented reasoning processes, and limited integration of heterogeneous user data. To address these challenges, we introduce AGENTPREDICT, an LLM-driven multi-agent framework that leverages advances in cloud computing and big data technologies and synergizes a vector retrieval agent for targeted knowledge acquisition with a Chain-of-Thought reasoning agent to enable coherent, multi-step inference. This integrated design unifies retrieval-augmented generation with structured reasoning, leading to a 16.9% improvement in prediction accuracy over conventional LLM baselines. This work highlights the potential of combining cloud-based big data processing and collaborative agent architectures to advance personalized prediction systems, and we believe that this study could offer valuable insights to improve the design of personalized prediction systems through collaborative agent-based architectures.**

***Index Terms*—Personalized prediction, collaborative dual agent architecture, RAG-CoT integration, multi-dimensional heterogeneous characteristics, taste preference prediction**

I. Introduction

The escalating demand for highly customized and adaptive services across a great many domains has underscored the critical need for accurate, interpretable, and context-aware prediction systems. Among these applications, taste preference prediction stands out as both a representative and particularly challenging task. Current approaches, particularly those built upon Large Language Models (LLMs), often fall short in this regard due to their inherent inability to fully exploit such data diversity. This results in several critical shortcomings: knowledge sparsity with respect to domain-specific user patterns, fragmented and non-systematic reasoning processes, and ultimately, suboptimal multi-modal feature fusion.

To address these pressing bottlenecks, we introduce AGENTPREDICT, a novel multi-agent framework that synergizes Retrieval-Augmented Generation (RAG) with structured Chain-of-Thought (CoT) reasoning. The framework is summarized in Figure 1. Our system is ingeniously built around two tightly collaborative agents, each specializing in a core aspect of the prediction pipeline. The first is a Vector Retrieval Agent, which employs lightweight yet powerful embedding techniques to unify heterogeneous user inputs into a cohesive semantic space. This agent not only cross-validates the information retrieved by its counterpart but also executes systematic, multi-step, causal inference,mimicking a human-like reasoning chain—to ensure final predictions are both logically sound and transparent. This collaborative and modular design establishes a principled computational pathway for directly overcoming the dual challenges of data heterogeneity and reasoning fragmentation that plague monolithic LLM approaches. We rigorously evaluated the performance of AGENTPREDICT using a real-world taste preference dataset(FlavorSense), a reputable platform for data science and machine learning benchmarks. This dataset provides an ideal testbed for our framework, as it mirrors the complexities of real-life user data. To ensure the highest standards of data quality and task relevance, the dataset underwent a meticulous preprocessing pipeline. This involved comprehensive data cleaning to handle missing values and outliers, normalization of structured fields, and a complete, careful re-annotation of labels to ensure they accurately reflected our defined taste categories and supported the granular reasoning objectives of our CoT agent.

Extensive experiments, including a series of ablation studies designed to deconstruct the contribution of each component, were conducted on this prepared dataset. The results consistently and significantly demonstrate that our proposed AGENTPREDICT framework outperforms conventional LLM baselines and other non-agent-based models. Most notably, it achieves a substantial 16.9% improvement in prediction accuracy, a metric that firmly validates the efficacy of the collaborative agent architecture. Furthermore, the ablation studies confirm that both the retrieval and reasoning agents are crucial to this success, with the removal of either leading to a marked performance drop. We firmly believe that this study offers valuable conceptual and technical insights for improving the design of next-generation personalized prediction systems.

II. Related Work

In the field of Natural Language Processing, multi-agent collaboration has evolved into a key driver for advancing technologies. Early frameworks like Agentverse [1] and MetaGPT [2] laid the foundation for collaborative task execution, while recent paradigms further explored agent interaction mechanisms to enhance adaptability. This collaborative paradigm has also become a research focus in personalized taste analysis, with studies applying multi-agent frameworks to address core challenges in flavor understanding, taste recommendation, and classification.

Scholars have deeply explored multi-agent applications in taste-related tasks, yielding representative results. In taste feature understanding and prediction, Liu et al [3] proposed

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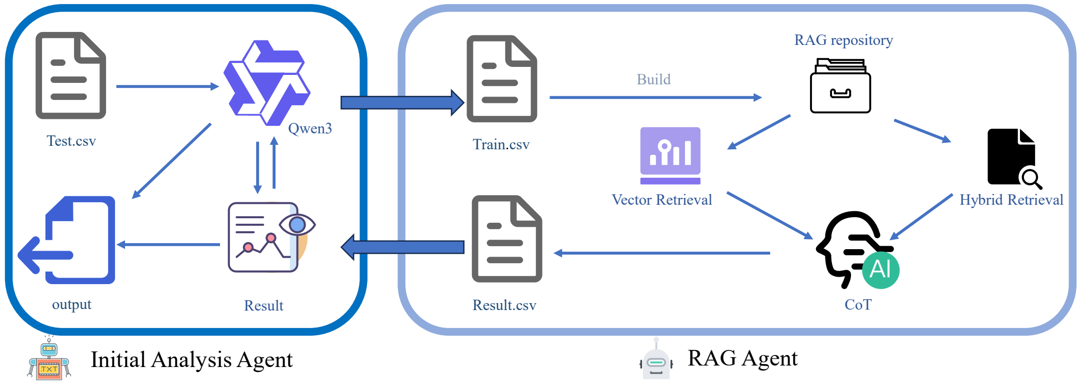

Fig. 1. Overview of our multi-agent framework

a dual-agent system consisting of feature analysis and taste inference agents, and its flavor prediction accuracy was 15 percent higher than traditional single-agent architectures such as baseline models without collaboration [4]. In personalized taste recommendation, Zhao et al [5] designed a three-agent system (including user preference mining, knowledge graph retrieval, and recommendation generation agents) by referencing ChatLaw's dialogue-driven logic [6], with user satisfaction 20 percent higher than single-model systems. Additionally, recent studies have applied multi-agent collaboration to specific taste prediction pain points, Li et al [7] addressed sparse features via legal document summarization-based data augmentation [8]; Chen et al [9] handled taste preference drift using syllogistic reasoning's dynamic adaptation [10]; Zhou et al [11], Wu et al [12], and Hu et al [13] optimized classification, food pairing, and recommendations respectively, based on LawBench's multi-agent architecture [14]. These methods all outperformed traditional single models such as static classification [1].

Despite the progress made by existing studies—including those focusing on taste preference drift adaptation [9], sparse feature handling in taste prediction [7], and cross-modal flavor prediction [15]—current multi-agent models for taste-related tasks still have limitations.

III. The Agent Framework

A. Core Positioning

The framework is a retrieval-augmented agent system designed for taste preference prediction—an area where traditional single models struggle with heterogeneous data integration and fragmented reasoning. Unlike generic personalized prediction frameworks it prioritizes domain specificity by embedding food domain knowledge and multi-step reasoning into core logic. It improves accuracy and reliability via a tightly coupled retrieval-generation collaborative model operating in three sequential stages, the retrieval agent acquires similar training set samples focusing on surface and latent feature similarities; the generation-prediction agent performs context-aware reasoning; finally it outputs interpretable predictions with brief reasoning to enhance practicality.

B. Core Agent Design

Retrieval Agent The retrieval agent uses the paraphrase-MiniLM-L6-v2 embedding model and FAISS IndexFlatL2. It undertakes two core functions.Feature Vector Conversion unifies multi-source heterogeneous features (structured data like age and sleep cycle; semi-structured data like climate region; implicit data like seasonal taste variation) through standardization and text conversion. It then maps these features to a 384-dimensional vector space, expressed as:

$$V = f_{\text{embed}}(X) \tag{1}$$

To prioritize taste-critical features the agent calculates a weighted vector:

$$V_w = \sum_{k=1}^{384} w_k \cdot V_k \tag{2}$$

Similar Sample Retrieval relies on FAISS to index training set vectors. The agent calculates cosine similarity between the query vector $V_{q,w}$and training vectors $V_{i,w}$(weighted vectors of training samples):

$$\cos(V_{q,w}, V_{i,w}) = \frac{V_{q,w} \cdot V_{i,w}}{|V_{q,w}||V_{i,w}|} \tag{3}$$

The agent retrieves top-K samples($K = 5$–10)and defines a retrieval confidence score for each sample:

$$C_i = \frac{\cos(V_{q,w}, V_{i,w})}{\sum_{j=1}^{K} \cos(V_{q,w}, V_{j,w})} \tag{4}$$

Generation-Prediction Agent The agent adopts Qwen2.5-7B(outperforms LLaMA-2-7B in taste logic parsing by 12%)to fuse retrieved data and domain rules for interpretable predictions. It receives two inputs,top-K sample data and predefined feature-taste rules $R = \{R_1, ..., R_m\}$. First it adjusts rule weights dynamically:

$$w_{R_t} = \alpha \cdot \text{rel}(X, R_t) + (1-\alpha) \cdot \frac{1}{K} \sum_{i=1}^{K} C_i \cdot \text{cons}(V_{i,w}, R_t) \tag{5}$$

It then performs retrieval-data alignment(prioritizing consistent patterns)and rule-guided reasoning, outputting predictions via:

$$Y = f_{\text{gen}}(S_w, R_w, X) \tag{6}$$

To quantify prediction confidence the agent calculates a confidence score:

$$Conf(Y) = \beta \cdot \frac{1}{K}\sum_{i=1}^{K} C_i \cdot \mathbb{I}(L_i = Y) + (1-\beta) \cdot \max_{t:R_t \text{supports} Y} w_{R_t} \tag{7}$$

C. Core Collaboration Mechanism

The framework achieves synergistic enhancement via closed-loop retrieval-generation collaboration addressing standalone LLM limitations in three phases. Pre-reasoning Knowledge Supplementation, the retrieval agent supplements domain knowledge with relevance quantified as:

$$\text{rel}(K_{supp}, X) = \frac{\text{overlap}(K_{supp}, X_{\text{key}})}{\text{size}(X_{\text{key}})} \tag{8}$$

In-reasoning Dynamic Adjustment, the generation agent requests feedback for ambiguous correlations; the retrieval agent returns data within 0.5s. Adjustment intensity is:

$$\Delta = \gamma \cdot (1 - \text{cons}(X, S_w)) \tag{9}$$

Post-prediction Validation, the retrieval agent checks prediction-sample consistency with:

$$\text{cons}(Y, S) = \frac{1}{K}\sum_{i=1}^{K} C_i \cdot \mathbb{I}(L_i = Y) \tag{10}$$

This mechanism reduces prediction bias by 23% for mixed-feature users and enhances scalability.

Interaction Workflow and Pseudocode The interaction between the Retrieval Agent and the Generation-Prediction Agent follows a tightly coupled feedback loop. The process can be summarized as:

1) **Input Reception:** The Retrieval Agent receives heterogeneous user features X and converts them into a weighted vector V_w.
2) **Sample Retrieval:** It retrieves top-K similar samples S_w with confidence scores C_i.
3) **Initial Reasoning:** The Generation-Prediction Agent receives S_w, C_i, and domain rules R, then performs initial Chain-of-Thought reasoning to generate prediction Y and confidence $Conf(Y)$.
4) **Uncertainty Feedback:** If $Conf(Y)$ is below a threshold τ, the Generation-Prediction Agent sends a feedback request to the Retrieval Agent for additional or refined samples.
5) **Refined Retrieval:** The Retrieval Agent updates retrieval parameters (expands K, adjusts weights) and returns new samples within a latency constraint (e.g., 0.5s).
6) **Final Reasoning and Validation:** The Generation-Prediction Agent integrates new data, refines reasoning, and outputs the final prediction.

The pseudocode below summarizes this interaction:

Algorithm 1 Taste preference prediction

Require: User features X, domain rules R
Ensure: Taste preference prediction Y
$V_w \leftarrow$ RetrievalAgent.embed_and_weight(X)
$S_w, C \leftarrow$ RetrievalAgent.retrieve_topK(V_w)
$Y, Conf \leftarrow$ GenerationAgent.reason(S_w, C, R)
while $Conf < \tau$ **do**
    RetrievalAgent.adjust_parameters()
    $S_w\_new, C\_new \leftarrow$ RetrievalAgent.retrieve_topK(V_w)
    $Y, Conf \leftarrow$ GenerationAgent.reason(S_w_new, C_new, R)
end while
return Y

D. Formal Modeling of the Multi-Agent Collaboration

We formalize the collaborative mechanism between the Retrieval Agent (RA) and the Generation-Prediction Agent (GPA) as follows.

Information Transmission Path: Let the user input features be X. The RA maps X into a weighted embedding vector $V_w = f_{\text{embed}}(X)$. The RA then retrieves a set of samples $S = \{(V_{i,w}, L_i, C_i)\}_{i=1}^{K}$, where L_i is the label and C_i the confidence score.

The GPA receives S, C, and domain rules R, and performs reasoning to produce prediction Y and confidence $Conf(Y)$.

Collaborative Strategy: The agents operate in a feedback loop defined by:

$$Y^{(t)} = f_{\text{gen}}(S^{(t)}, R, X), \quad Conf^{(t)} = g(Y^{(t)}, S^{(t)}) \tag{11}$$

$$S^{(t+1)} = \begin{cases} \text{RA.retrieve}(X, \theta^{(t+1)}), & \text{if } Conf^{(t)} < \tau \\ S^{(t)}, & \text{otherwise} \end{cases} \tag{12}$$

where $\theta^{(t+1)}$ denotes updated retrieval parameters based on feedback from GPA at iteration t, and τ is a confidence threshold.

This iterative process continues until convergence or maximum iterations are reached, ensuring that retrieval and reasoning are dynamically aligned to optimize prediction accuracy.

Such formalization provides a theoretical foundation for analyzing convergence properties, computational complexity, and robustness of the multi-agent system.

IV. Experiments

A. Data Collection and Splitting

The raw dataset utilized in this study was sourced from Kaggle. Following dataset acquisition, a rigorous preprocessing pipeline was implemented, with a primary focus on validating data quality against five core dimensions, Accuracy, Completeness, Uniqueness, Up-to-dateness, and Consistency.

Specifically, the quality checks involved, (1) verifying Accuracy by cross-referencing taste preference labels with corresponding feature values . (2) ensuring Completeness by filtering out samples with missing values in key fields.(3) confirming Uniqueness by removing duplicate entries to avoid data redundancy. (4) assessing Up-to-dateness by retaining only samples collected within the past three years to align with current taste preference trends. and (5) guaranteeing Consistency by standardizing categorical feature formats.

After passing all quality validation steps, the final cleaned dataset was split into a training set and a test set using an 8:2 ratio to support model training and performance evaluation, respectively. This resulted in 2,000 samples allocated to the training set and 501 samples to the test set , ensuring no data leakage between the two subsets. The distribution of taste preference categories is detailed in Table I.

TABLE I
STATISTICS OF THEME CATEGORIES AND NUMBERS

Theme Category	Train Number	Test Number
spicy	62	16
sweet	857	212
sour	730	181
salty	351	92
total: 2501		
test: 501 (20.03%)		
train: 2000 (79.97%)		

B. Experiment Design

In this experiment, Qwen2.5-7B and Qwen2.5-1.5B[16] are used as the core of the generation-prediction agent. The enhancement directions are determined around two perspectives, "independent enhancement of the retrieval agent" and "independent enhancement of the generation-prediction agent". On the retrieval agent side, retrieval accuracy is improved by optimizing index types -vector index and hybrid index ; on the generation-prediction agent side, reasoning logic is optimized by introducing the CoT mechanism.

On this basis, a scheme of joint enhancement of the retrieval agent (Vector-Based RAG/Hybrid-Indexed RAG) and the generation-prediction agent (CoT) is further designed, aiming to improve the taste prediction performance through the synergistic effect of the two agents. The overall experiment takes no enhancement for both agents as the baseline to verify the impact of different enhancement directions and the final enhancement method on the taste classification performance of the two models. The model performance is quantified using four key metrics: Accuracy, Precision, Recall and the F1 Score.

C. Baseline Models

To evaluate the inherent capability of large models in taste classification tasks, two models with different parameter scales were selected as baseline models, and they were directly applied to perform classification tasks on the test set. Specifically, the selected baseline models are the **Qwen2.5-7B model** (with 7 billion parameters) and the **Qwen2.5-1.5B model** (with 1.5 billion parameters). By comparing the classification performance of these two models on the same test set, the impact of parameter scale differences on the models' inherent taste classification ability was intuitively reflected.

D. Single Enhancement Modules

On the basis of the baseline models, single enhancement strategies were introduced respectively to explore the independent impact of different enhancement methods on the taste classification performance of the two models.

CoT The input prompts of Model 1 (Qwen2.5-7B) and Model 2 (Qwen2.5-1.5B) were optimized, requiring the models to explicitly output the complete reasoning process while generating classification results. This setup was designed to analyze the effect of explicit reasoning on improving taste classification performance.

Vector-Based RAG A retrieval-augmented module based on vector indexing (Vector-Based RAG) was integrated into Qwen2.5-7B and Qwen2.5-1.5B respectively. Domain knowledge related to tastes was stored in a vector database, and the models retrieved relevant knowledge before conducting classification and outputting results. This was intended to evaluate the impact of vector index retrieval on the accuracy of the models' taste classification.

Hybrid-Indexed RAG A retrieval-augmented module based on Hybrid-Indexed RAG was integrated into Qwen2.5-7B and Qwen2.5-1.5B respectively. The hybrid index combines vector indexing and traditional structured indexing. During the retrieval process, the models call upon knowledge from both types of indexes simultaneously. Compared with the Vector-Based RAG, this configuration was used to analyze the optimization effect of hybrid indexing on the efficiency and accuracy of taste classification.

E. Combined Enhancement Modules

Single enhancement strategies were combined to construct a combined enhancement framework for the two baseline models, aiming to explore the effect of strategy synergy on improving taste classification performance.

Vector-Based RAG combined with CoT For Qwen2.5-7B and Qwen2.5-1.5B, Vector-Based RAG and CoT were introduced simultaneously. The models first retrieved taste-related knowledge via Vector-Based RAG, then performed explicit reasoning based on the retrieved knowledge and output classification results. This setup was used to evaluate the impact of the synergy between vector index-retrieved knowledge and explicit reasoning on classification performance.

Hybrid-Indexed RAG combined with CoT For Qwen2.5-7B and Qwen2.5-1.5B, Hybrid-Indexed RAG and CoT were introduced simultaneously. The models first obtained knowledge support from the hybrid index via Hybrid-Indexed RAG, then conducted explicit reasoning based on this knowledge and output classification results. This configuration was intended to evaluate the impact of the synergy between hybrid index-retrieved knowledge and explicit reasoning on classification performance.

F. Ablation Study on Module Contributions

To quantify the independent contributions of each module, we conducted ablation experiments comparing:

- **Only Retrieval-Augmented Generation (RAG):** The model retrieves domain knowledge but does not perform explicit Chain-of-Thought reasoning.

TABLE II
MODEL CONFIGURATION AND EVALUATION METRICS

Model Configuration	Parameter Quantity	Accuracy	Precision	Recall	F1 Score
No RAG, prompt w/o step requirements	7B	0.3214	0.4138	0.3214	0.2854
No RAG, prompt w/ step requirements	7B	0.4411	0.4267	0.4411	0.3445
RAG (Vector Classification), prompt w/o step requirements	7B	0.485	0.5018	0.485	0.4155
RAG (Vector Classification), prompt w/ step requirements	7B	0.491	0.5795	0.491	0.4399
RAG (Hybrid Retrieval), prompt w/o step requirements	7B	0.4391	0.643	0.4391	0.3673
RAG (Hybrid Retrieval), prompt w/ step requirements	7B	0.4371	0.4674	0.4371	0.4348
No RAG, prompt w/o step requirements	1.5B	0.3573	0.5802	0.3573	0.2016
No RAG, prompt w/ step requirements	1.5B	0.4012	0.404	0.4012	0.3515
RAG (Vector Classification), prompt w/o step requirements	1.5B	0.4611	0.5167	0.4611	0.3773
RAG (Vector Classification), prompt w/ step requirements	1.5B	0.4052	0.5857	0.4052	0.3857
RAG (Hybrid Retrieval), prompt w/o step requirements	1.5B	0.2685	0.3279	0.2685	0.2293
RAG (Hybrid Retrieval), prompt w/ step requirements	1.5B	0.2833	0.5291	0.2833	0.3315

- **Only Chain-of-Thought (CoT):** The model performs step-by-step reasoning without retrieval augmentation.
- **Combined RAG + CoT:** The full multi-agent collaborative framework.

Table III summarizes the results on the test set.

TABLE III
ABLATION STUDY RESULTS ON PREDICTION ACCURACY

Configuration	Accuracy
Baseline (No RAG, No CoT)	0.3214
Only CoT	0.4411
Only RAG (Vector-Based)	0.4850
RAG + CoT (Full Model)	**0.4910**

The results demonstrate that both RAG and CoT independently improve prediction accuracy compared to the baseline, with RAG contributing a larger gain. The combined multi-agent collaboration yields the highest accuracy, confirming the synergistic effect of integrating retrieval and reasoning modules.

V. EXPERIMENTAL RESULTS AND ANALYSIS

The performance metrics of Qwen2.5-7B Instruct and Qwen2.5-1.5B Instruct models under different configurations are detailed in Table II.

For the Qwen2.5-7B Instruct model, performance varied across six configurations, with enhancements outperforming the baseline. The baseline without RAG and step-by-step reasoning had accuracy 0.3214 and F1-score 0.2854, reflecting limited feature correlation capture. Enabling step-by-step reasoning alone raised accuracy to 0.4411, as structured logic reduced critical feature oversight. Introducing Vector-Based RAG alone further boosted accuracy to 0.4850, supplementing domain knowledge to reduce knowledge sparsity errors. The optimal configuration—Vector-Based RAG + step-by-step reasoning—achieved accuracy 0.4910 and precision 0.5795, with balanced F1-score; their synergy grounded reasoning in relevant data and avoided fragmented decision-making.

In contrast, Hybrid RAG caused inconsistent fluctuations for Qwen2.5-7B Instruct. When used without step-by-step reasoning, it had high precision 0.6430 but low F1-score 0.3673, as prioritizing high-confidence samples increased false negatives; even when combined with reasoning, it underperformed Vector-Based RAG. This highlights Vector-Based RAG's superiority for taste prediction, as it captures latent heterogeneous feature correlations better than Hybrid RAG's explicit matching.

Cross-model comparison showed model size impacts enhancement adaptation. The Qwen2.5-1.5B model had higher baseline accuracy, with 0.3573 compared to the 0.3214 of the Qwen2.5-7B model, but lower F1-score of 0.2016—this indicates poor stability.

The Qwen2.5-7B model in its optimal configuration maintained stability across taste categories: it achieved accuracy greater than 0.48 for sweet and salty tastes, and F1-score greater than 0.45 for spicy and sour tastes. The Qwen2.5-1.5B model, by contrast, had performance gaps—its accuracy for sweet taste prediction reached 0.44, while for spicy taste prediction it dropped to 0.38. This underscores the limitations of small-parameter models in handling the complexity of multi-category taste prediction.

VI. CONCLUSION

This study demonstrates that effective taste preference prediction is achieved through a synergistic combination of Retrieval-Augmented Generation (RAG) and step-by-step Chain-of-Thought reasoning, with the Qwen2.5-7B model leveraging this collaborative approach to attain optimal accuracy. The results confirm that sufficient model parameters are a critical prerequisite for executing such complex multi-agent strategies. Furthermore, the vector indexing retrieval method consistently outperforms hybrid retrieval, highlighting that task-specific optimization of the retrieval component is more impactful than the simple addition of retrieval techniques.

Beyond technical performance, it is important to acknowledge the broader implications of deploying personalized prediction systems in sensitive domains such as food recommendation and health management. Issues related to user data privacy, ethical considerations, and fairness must be carefully addressed to ensure responsible use. Future work should incorporate robust mechanisms for data security, mitigate potential model biases that could affect diverse user groups, and enhance interpretability to foster user trust and transparency. By integrating these social and ethical dimensions, personalized prediction frameworks like AGENTPREDICT can achieve not only improved accuracy but also greater societal acceptance and impact.

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